

EE 2080: Random Processes

Programming Assignment: Sample Questions

1 Random Process: Basics

1.1 Random-Phase Sinusoid Process

Consider the random-phase sinusoid process $\{X_n = \cos(2\pi * 0.1n + \Theta), n \in \mathbb{N}\}$, where $\Theta \sim \text{unif}[0, 2\pi]$. Generate and plot on the same figure 5 sample paths of the process for $n \in \{1, 2, \dots, 50\}$.

Remark: This random process is frequently used to model an analog sinusoid whose phase is unknown and that has been sampled by an analog-to-digital converter.

1.2 White Gaussian Noise (WGN) Process

A common model for physical noise, such as resistor noise due to electron motion fluctuations in an electric field, is termed white Gaussian noise (WGN). It is assumed that the noise has been sampled in time to yield a discrete-time continuous-valued random process $\{X_n : n \in \mathbb{N}\}$. The WGN random process is defined to be the IID process where $X_n \sim \mathcal{N}(0, \sigma^2)$, $n \in \mathbb{N}$.

Generate an overlaid scatterplot (`matplotlib.pyplot.scatter`) of 50 realizations of the WGN process with variance 1.

1.3 Moving Average Process

Consider a WGN process $\{X_n : n \in \mathbb{N}\}$ with variance 1. We define the moving average process $\{Y_n, n \in \mathbb{N}\}$, where $Y_n = \frac{1}{N} \sum_{i=0}^{N-1} X_{n-i}$. Simulate $M = 10^4$ realizations of the MA process for $n \in \{1, 2, \dots, 10\}$. Using these realizations,

- estimate and plot the mean process, $\mu_Y(n)$, for $n \in \{1, 2, \dots, 10\}$
- estimate the covariance $C_Y(\Delta) = \frac{1}{M} \sum_{i=1}^M y_{n_1}^i y_{n_2}^i$, where y_n^i is the realization of Y_n in the i^{th} sample path.

Using these mean and covariance estimates, determine if $\{Y_n, n \in \mathbb{N}\}$ is a W.S.S process.

2 Gambler's Ruin

2.1 Vanilla Version

A gambler starts with wealth k dollars and has a target $n > k$. He bets \$1 in each round and wins each round with probability p . Let X_n denote the gambler's wealth at time step n , $n \geq 0$ with $X_0 = k$. Let the stopping time for this process $\{X_n : n \geq 0\}$ be defined as the time step at which the gambler wins or is ruined (whichever comes first). For a given (k, n, p) triple,

- Plot five sample paths till the gambler reaches the target or is ruined (whichever comes first).
- Using Monte Carlo simulation over 10^4 trials, estimate the probability that the gambler wins (reaches the target before being ruined).
- Plot the value of the stopping time for 100 of the iterations in which the gambler wins and 100 iterations in which the gambler is ruined.

Default value of $(k, n, p) = (40, 60, 0.5)$.

2.2 Multiple-stake version: Uniform Betting Strategy

A gambler starts with wealth k dollars and has a target $n > k$. At each step t he chooses a stake b_t uniformly at random from $\{1, 2\}$, subject to affordability ($b_t \leq$ his current wealth). If he bets \$1, he wins with probability p_1 and if he bets \$2, he wins with probability p_2 .

- On a win, his wealth increases by b_t , i.e., $W_{t+1} = \min(W_t + b_t, n)$
- On a loss, his wealth decreases by b_t , i.e., $W_{t+1} = \max(W_t - b_t, 0)$

For a given (k, n, p_1, p_2) -tuple, simulate sample paths till win/ruin and estimate win probability using Monte Carlo simulation over 10^4 trials. Also, plot the value of the stopping time for 100 of the win iterations and 100 iterations in which the gambler is ruined.

Default value of $(k, n, p_1, p_2) = (40, 60, 0.6, 0.4)$

2.3 Multiple-stakes: Comparison of betting policies

A gambler starts with wealth k dollars and has a target $n > k$. At each step t he chooses a stake b_t from $\{1, 2\}$, subject to affordability ($b_t \leq$ his current wealth), according to a given betting policy. If he bets \$1, he wins with probability p_1 and if he bets \$2, he wins with probability p_2 . For a given (k, n, p_1, p_2) -tuple, consider the following betting strategies:

- Always-1: Bet 1\$ always
- Always-2: Bet 2\$ whenever possible, if wealth = \$1, bet \$1.
- Pessimistic: If $p_i > p_j$, choose the Always-i strategy
- Uniform-Random: choose uniformly among admissible bets $\{1\}$ or $\{1, 2\}$
- Threshold: Bet \$2 if wealth $W_t \geq T$, else bet \$1. Set the threshold T at 10\$ as default.
- Adaptive: $b_1 = 1$. If win in round t , $b_{t+1} = 2$, If loss in round t , $b_{t+1} = 1$.

Tasks:

- Plot two sample paths for each betting policy
- Histogram of stopping time per policy
- Threshold sweep: Sweep $T \in \{2, 4, 6, \dots, 20\}$ and plot the success probability vs T and the average stopping time vs T .
- For a given (k, n, p_1, p_2) -tuple and the default value of T , output a table with the estimated success probabilities and average stopping-times of all betting policies and declare the best policy w.r.t success probability.

Default value of $(k, n, p_1, p_2) = (40, 60, 0.6, 0.4)$